NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mathematics and Computing Technology

Mid-Semester Examination - October 2025

Course: Engineering Mathematics — Code: MC011201/ MC011301

Time: 2 hour — Maximum Marks: $6 \times 5 = 30$

Answer all questions

1. Use elementary operation to obtain the inverse of the matrix

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{pmatrix}.$$

2. Determine the values of k for which the linear system of equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ x_1 + 2x_2 + 4x_3 &= k \\ x_1 + 4x_2 + 10x_3 &= k^2. \end{aligned}$$

has (i) a unique solution; (ii) infinitely many solutions and (iii) no solution.

3. Test that the vectors $u_1 = (2, -3, 1)$, $u_2 = (1, 4, -2)$, $u_3 = (-8, 12, -4)$, $u_4 = (1, 37, -17)$, and $u_5 = (-3, -5, 8)$ generates $\mathbb{R}^3$. Find the subset of the set $\{u_1, u_2, u_3, u_4, u_5\}$ that is a basis for $\mathbb{R}^3$.

4. Let $F : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear mapping defined by

$$F(x, y, z, t) = (x - y + z + t, 2x - 2y + 3z + 4t, 3x - 3y + 4z + 5t).$$

Find a basis and dimension of the image and kernel of F.

5. Consider the following bases of $\mathbb{R}^2$:

$$S = \{u_1, u_2\} = \{(1, -2), (3, -4)\} \quad \text{and} \quad S' = \{v_1, v_2\} = \{(1, 3), (3, 8)\}.$$

   (a) Find the coordinate of $v = (a, b)$ relative to the basis S.

   (b) Find the change-of-basis matrix P from S to S'.

   (c) Find the coordinates of $v = (a, b)$ relative to the basis S'.

   (d) Find the change-of-basis matrix Q from S' back to S.

6. Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

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